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Interviewer

2 Could you please briefly introduce yourself and provide a bit of background on your experience working with genomic data?

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Participant

4 Yeah, I'm a graduate researcher in the Biomedical Informatics and Medical Education Program in the Medical School of the University of Washington. I'm currently entering third year, and I used to be a bioinformatics programmer at another institute finishing my master's degree. I had kind of like a prior one year experiences before that. So for now, like, about two years or three years of experience.

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Interviewer

6 And what kind of experience have you had working with genomic data?

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Participant

8 Before? It was mostly small sample size, VCF file, mutation data set from the exome. For now, it's more about population studies. So we have more data sets enriched in the population level and also mostly just mutation files at VCF formats.

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Interviewer

10 Do you remember about your first experiences working with genomic data? How did you get started?

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Participant

Yeah, I do remember. I remember the learning curve is so hard. I get started from the request of my previous Pi in my master degree to help with another project running, like, some of the pipeline through the mutation calling in the VCF file. And honestly, that was terrible for a starter because a lot of formats of the file is not really well documented. For example, VCF file, you could have a lot of different headers and annotations in those columns. And also the pipeline, usually it's in Bash, and some of them, they have like, version control, different softwares you use in Bash to run the pipeline. That also brings you a lot of choices, but also for a starter, the documentation and the reason you run through the

2:1 I remember the learning curve is so ...
First experiences
2:3 honestly, that was terri...
Cognitive reaction
Emotional reaction

12 pipeline is it's really hard, like, for start to figure out every component in that. So that's going to cause a lot of headache.

2:1 2:3

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Interviewer

14 That makes sense. Did you have any educational resources you use to help you out?

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24 **Participant**

16 Yeah, I do, I think, but most of them are from the lab members. Like, they give you some of some notes or previous people published websites and their documentation of their analysis. So it's not like officially courses or curriculum things. It's almost just documentation from other people.

2:2 most of th... Resources used w...

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Interviewer

18 From the lab that you were in.

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Participant

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20 The lab or others. Yeah.

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27 **Interviewer**

Got you. Currently and in the past, what kinds of genomic analysis software have you utilized to conduct your research? I know you mentioned some Bash scripts, so is there any specific software?

Participant

So I think my experience is mostly, I think some of the Bash command line software, like Sam tools before and also the Alignment tools like PWA or something. I don't remember that. One of the RNA alignment tool I think is called Selfish or something. Selfish for now, let me think. Oh, yeah. And for now it's more populational level. So I also use like, Plink.

2:4 I think my exp... Specific software t...

Interviewer

And when you're using these tools currently, like Plink and other tools, what is your workflow or pipeline look like?

Participant

28 Yeah, I think that's a really good question because I remember when I started using Plink, there's another colleague who had more experiences who basically wait, I think I did. Could you repeat the question?

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Interviewer

30 So the question was what kind of workflow or how are you using your software to perform your analysis?

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Participant

Oh, yeah, I think we were talking about Plink. So one of the colleagues, I think before I met, he had a bunch more experiences specifically running scripts, the current server I'm using. So he was using the upstream some script called Perl. I think I got time to like reformatting some of the files from the VCF and to fit them sometimes into a linear model. I think that
33 yeah, that experience was also, you know, purely just based on other people's experiences. It's not very flexible or you can read it from somewhere. And when I was looking at the menu of plink, I didn't
34 understand that thing actually. How do I extract information from LD regions or stuff? The annotations are so bad. And really the workflow, I just go step by step and asking people, hey, can you help with this? Or do you have any serious working from that? So it's just asking a bunch of help from others.

2:5 It's not very flexible or ...
Barriers of specific...
Cognitive reaction

35 **Interviewer**

How do you feel when you're working with plink and similar software? I know you before you mentioned there was some frustration with it not being clear of how to use it, but are there other emotions that you feel when working with those kinds of analysis tools?

Participant

In fact, I would say mostly it's really frustrating in a way, but if I also consider it another way, like the data set is really big. But I also definitely appreciate it because without plink, I don't know how I'm going to deal with running those huge analysis using a lot of resources. Yeah, I definitely appreciate that way because it looks like plink is really effective and they take care of a lot of things, but actually in the process, yeah, it definitely caused a lot of frustrations because of their documentation and some of those error message. It's not really clear when you also Google that there are a bunch of other people googling the same things you could find and the answers are actually various ways and some people find some of the

2:6 In fact, I would say mostly it's really f...
Cognitive reaction
Emotional reaction

36 answers for somewhere. I think it's really frustrating when you're doing that. Sometimes it just don't work.

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Interviewer

38 Did you have any questions or concerns when you started using it?

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47 **Participant**

40 Let me see. I think mostly it would be their documentation and I would
48 specifically say their example scripts. I think so as long as they listed all the hashtag and the functions. Right. I would definitely wanted them to provide more examples of the workflow and how the tutorial would go. Like, if you were doing some genomic analysis and you need to say grab or filter specific regions, what do you do and what do you most of the frequently asked questions or errors, I think that'll be more helpful for this.

2:7 I would definit... Suggested improv...

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Interviewer

42 Okay, I think that wraps up a bit about our background. Let's move about let's move on to some positives and negatives and then some impact. So, thinking back, what do you see as positives about utilizing genomic software that you've utilized, such as Plink and some of the other ones?

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Participant

Yeah, the positive side is I really appreciate the efficiency. I think some of the remote clusters have more cores and rams. You can finish a job really, really quick, even dealing with large sample data sets. So that was so powerful in my opinion. So I definitely appreciate their integrations in their system and also their running speed, efficiency.

2:8 the positive si... Benefits of specifi...

Interviewer

What about some negatives about utilizing that genomic software?

Participant

Yeah, I think negative status, apart from the documentation, I would say some of the error message, which is not clear for debugging, and usually, even though we say it's pretty efficient, but you'd have a lot of lab members sometimes also running it in parallel so you won't be able to use exclusively. So sometimes, for example, if your job take an hour or two or three to finish it and give you a really blurred or unreadable error message.

2:9 I think negative sta... Barriers of specific...

48 For example, I have been getting like bust error, blah, blah, blah, dumped core or something. I don't really understand that. And they dumped a huge binary file in the system and no output. And when you Google that, there's a lot of different answers. Some says that it's a Ram issue, some says it's something like the version doesn't work in that way. So I definitely
55 would like them to provide a better information error message. And also if you're running like a super long script, that would be so painful, like you get yeah.

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Interviewer

50 It seems like it would be almost like a waste of time because you have to rerun it.

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Participant

52 Yeah. And you still don't figure out why.

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Interviewer

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59 Interesting. Has, like, utilizing plink in your environment influenced any, like, cost of your experiments? Has it increased, like how much you're spending for running the experiment or upkeeping the data?

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Participant

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Yeah, I think in general, it definitely promotes efficiency of working. So when we were talking about the learning curve right. I definitely see it's really rough to get started, but once you go through the pain and learn from those errors, gradually you figure out, okay, that's the way you work. Then it really gets more efficient.

2:10 I think in gen...

Impact on cost/up...

Interviewer

Got you.

Participant

Yeah, so I definitely appreciate that, but it's just so painful when you start.

Interviewer

Yeah, okay, that makes sense. That makes sense. If you were to do your analysis process all over again, would you utilize the same software or would you utilize something else?

Participant

I'm kind of lazy person, so if I learn the tool and I know how to use it, I would definitely just replicate and use the same. I know there are other choices. Like probably when you were talking with other colleagues, they would say, oh, look at that, there's a bunch of new software and people came out, they're adding more functions, and now you can do a more complicated model. But usually I was just, yeah, I'll stick with the old one that's employed.

Interviewer

That makes sense.

Participant

Stick with what you're doing. Yeah.

Interviewer

Okay. I know we talked about it briefly, but do you have any other recommendations for changes or additions to the software that would aid in your research? I know we already talked about better error messages and then more examples and better documentation, but is there anything else that you would like to change about the software?

Participant

Anything else that I could change?

Interviewer

Like, I know one of the things that I thought about when you were talking about running hour long scripts is if the script has portions that are unrelated or maybe they can give you some of the results so you can kind of have a feel for what's going on without having to wait the whole 3 hours.

Participant

76 Yeah, I definitely agree. You mentioned a really nice thing, I think, to just do write the intermediate results in some way in one way to show the progress or other way to preserve some of the results you have. Yeah, that would be a lot of help. Not only saving time, but you can poke around
77 parts of the results. Yeah, that's really good. Other than that, I think just better documentation and examples.

2:11 I think, to just do ...
Suggested improv...

78 **Interviewer**

Okay. This last question is more of an exploratory question. As the NHGRI and other genomic research organizations are moving towards cloud computing, what challenges do you anticipate facing when we're running genetic research in the cloud?

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Participant

80 Running the cloud? I was thinking about two things. I think the first is the scheduling issue, I think I don't know how that works in the cloud, but for currently, sometime in the lab, if one of the colleagues using all the cores or unintentionally or something just didn't limit anything then that would cause troubles for the others sometimes to run the script since he's basically using all the resources. I don't know how that can be scheduled in the cloud. Maybe they have better scheduling system in the usage limiting or something. The second thing I would be thinking about is the version control. I think so. I would guess there are probably many other or different users. They probably prefer different versions of the package. I was wondering if loading that and running like would also cause potential issues if they maintain a lot of different environment with that like sometimes missed out.

2:12 Running the cloud? I was thinking about two thi...
Challenges/conce...

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Interviewer

82 Yeah, that makes sense. Yeah. Awesome. Well those were all my questions, thank you so much!